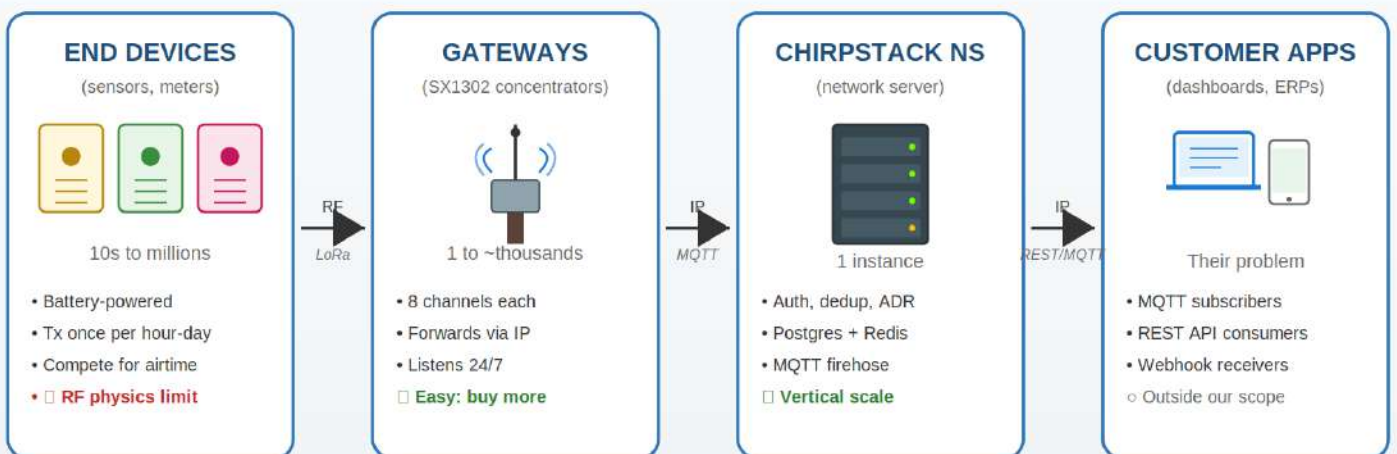


The Three-Layer Mental Model

Each layer has its own capacity limit — the one you hit first depends on your traffic pattern



LAYER 1 (RADIO) — Hardest to fix; binds the whole system

Channel collisions, duty cycle, spreading factor. Cannot be solved with more compute.

LAYER 2 (GATEWAY) — Buy more boxes

8 simultaneous channels per gateway, ~100M packets/day each

LAYER 3 (NS) — Vertical scale, then split components

Postgres + Redis + ChirpStack process. Easy to grow.

Spreading Factor: Range vs Airtime Trade-off

Every step up in SF doubles airtime but extends range — this is the central LoRaWAN trade-off

SF	Airtime (10-byte payload)	Typical range	Data rate
SF7 (short)	 ~46 ms	 2–5 km	5,470 bps
SF8	 ~82 ms	 3–7 km	3,125 bps
SF9	 ~165 ms	 5–10 km	1,760 bps
SF10	 ~330 ms	 7–15 km	980 bps
SF11		 10–18 km	440 bps
SF12 (longest)		 15–25+ km	250 bps

□ Why this matters for capacity:

Each gateway has ≈ 8 simultaneous channels. An SF12 packet (1.3s) occupies a channel $\sim 30\times$ longer than an SF7 packet (46 ms).

In the same airtime budget, one gateway can carry **$\sim 30\times$ more SF7 devices than SF12 devices**. Use ADR to keep devices at low SF.

Per-Gateway Capacity by Use Case

How many devices fit under one gateway depends entirely on how often they transmit



Quick capacity estimate per gateway

Rule of thumb formula:

$$\text{max_devices} \approx (3600 \times 8 \times 0.10) / (\text{uplinks_per_hour} \times \text{avg_airtime_seconds})$$

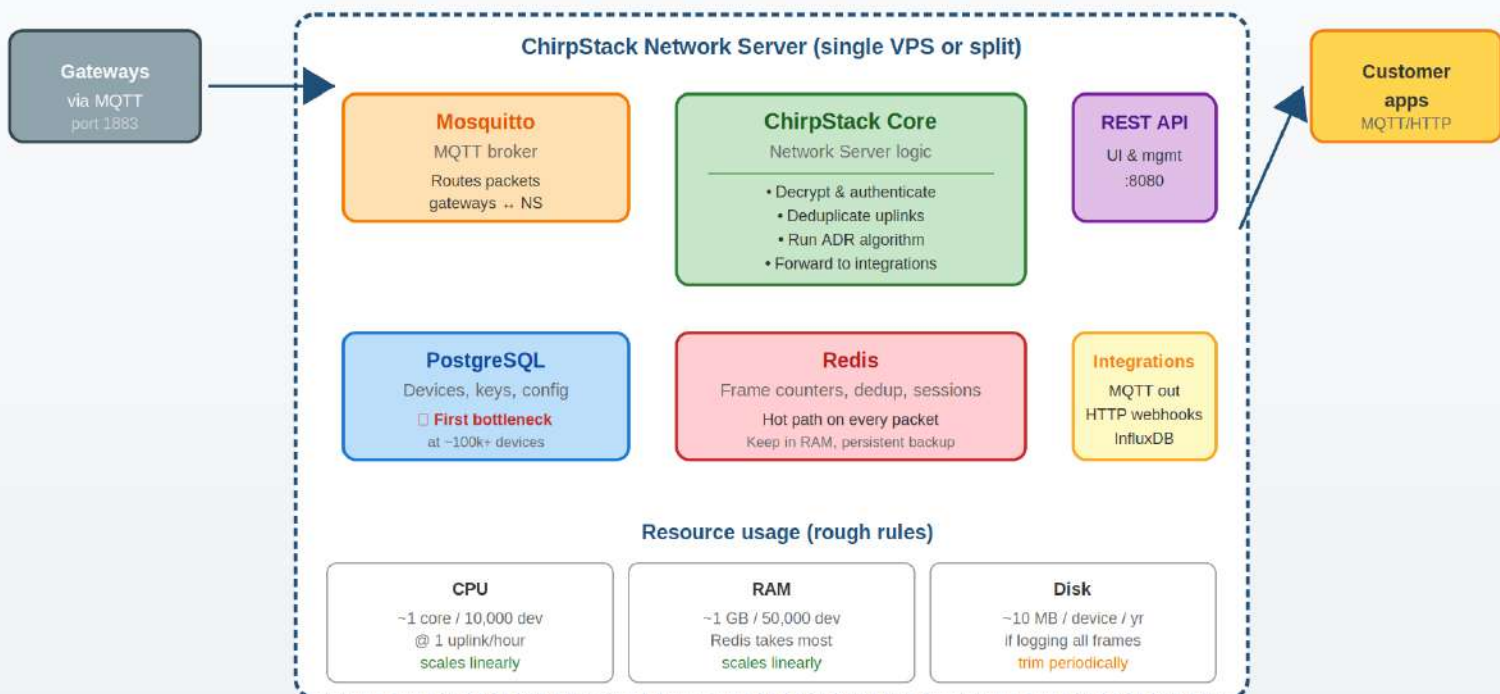
Example: 1 uplink/hour, SF8 (82 ms = 0.082 s) → max = 2880 / (1 × 0.082) = ~35,000 devices theoretical

Real world: halve for collisions, halve again for safety → ~8,000-10,000 in practice

The 0.10 factor = 10% channel utilisation. Above 20% collisions explode (ALOHA limit).

ChirpStack Network Server: Components & Bottlenecks

The five processes that handle every packet — and which one fails first at scale



Key insight

The radio layer fails long before the NS does. One €5/month VPS handles thousands of devices easily — but no compute can fix RF collisions.

VPS Sizing Tiers: From Laptop to Million Devices

What hardware do you need at each scale? Real numbers, real costs.

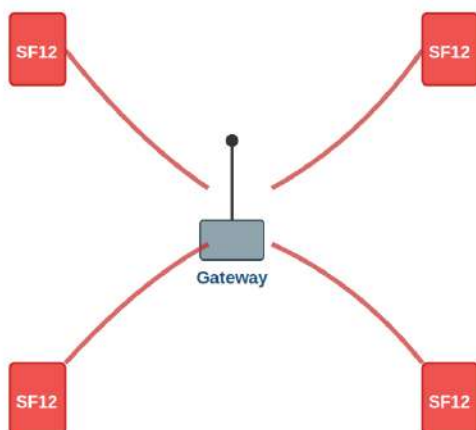
Tier 0: BENCH	Tier 1: PILOT	Tier 2: GROWTH	Tier 3: SCALE	Tier 4: HA
Laptop (where you are now)	Hetzner CX22 or DigitalOcean	Hetzner CX32 or equivalent	Dedicated server Hetzner AX/EX	Split + replicated Multiple servers
2 cores 4-8 GB RAM SSD any size	2 vCPU 4 GB RAM 40 GB SSD	4 vCPU 8 GB RAM 80 GB SSD	8-16 cores 32-64 GB RAM NVMe SSD RAID	3+ servers DB replica HA Redis
Capacity: • 1-3 gateways • ~100 devices • Demo only	Capacity: • 5-10 gateways • Up to 2,000 dev • First paying client	Capacity: • 30-50 gateways • Up to 20,000 dev • Multi-tenant	Capacity: • 200+ gateways • Up to 100,000 dev • Regional ops	Capacity: • 500+ gateways • 500,000+ dev • 99.9% SLA
Cost: \$0 Reuse hardware	Cost: ~€5/mo USD ~6/month	Cost: ~€10/mo USD ~12/month	Cost: ~€40/mo USD ~45/month	Cost: €200+/mo USD ~230/month
Good for: testing + 1st customer demo	Good for: pilot, 1-2 customers	Good for: 5-10 paying customers	Good for: WASAC, REG sized clients	Good for: national + regional operator

Growth path — each tier is a straightforward Postgres dump + restore migration

ADR (Adaptive Data Rate): Why It's the Most Important Lever

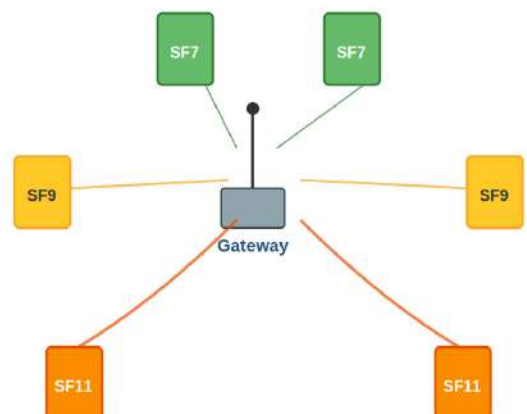
The network server tells each device to use the lowest SF that still works — multiplying capacity ~5×

Without ADR: every device at SF12



Each packet = 1.3 seconds airtime
Max devices/gateway: ~200

With ADR: each device at minimum SF

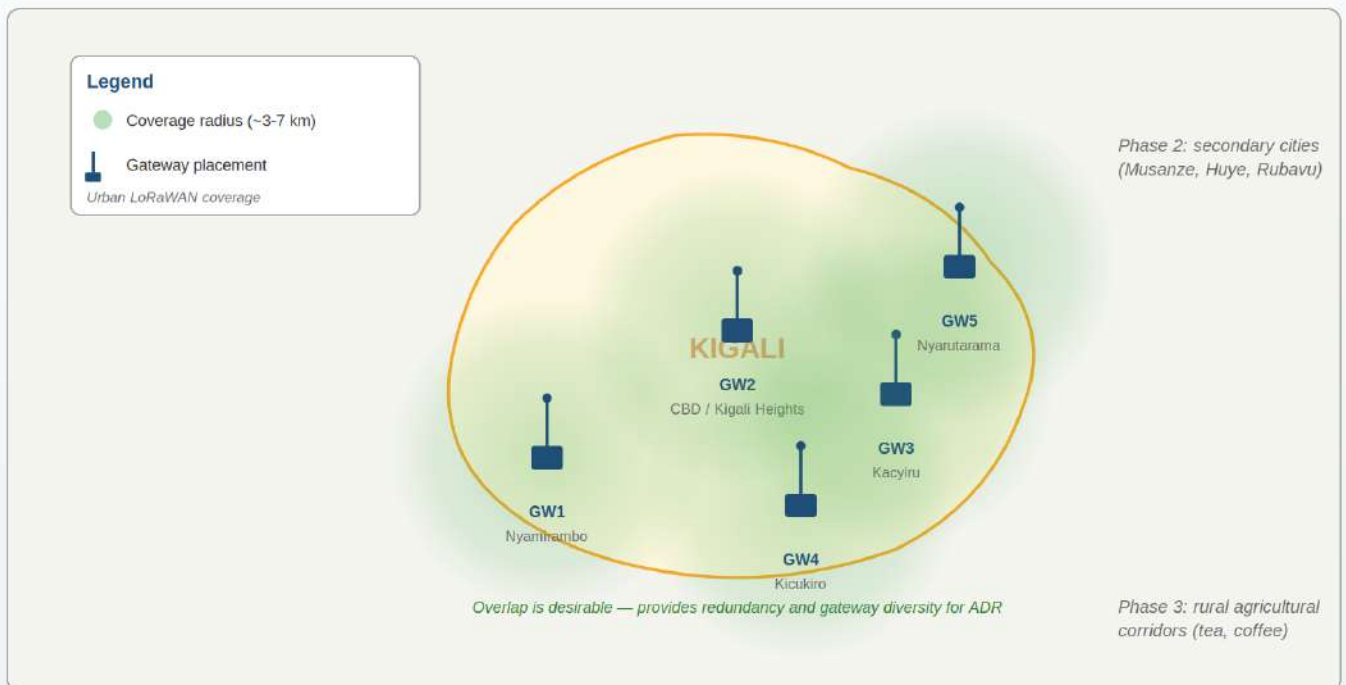


Avg packet = 150 ms airtime
Max devices/gateway: ~10,000

ADR is enabled by default on all device profiles have it on. It is the #1 capacity multiplier you don't pay for.

Gateway Deployment Strategy

Phase 1 = Kigali only. Each gateway covers ~3-7 km radius. Build density before geography.



Density-first principle:

5 gateways in Kigali alone serve more devices than 50 gateways spread thinly across the country. Cover one city completely before expanding to the next.